# Nash equilibrium

**Pierre Pinson**

Imperial College London, Dyson School of Design Engineering
**mail:** p.pinson@imperial.ac.uk - **webpage:** www.pierrepinson.com

**Game Theory and Mechanism Design** - Lecture 4

(This lecture heavily relies on Tadelis (2013) Game Theory – An Introduction – chap. 5, and Narahari (2014) Game Theory and Mechanism Design – chap. 6)

# Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

- Describe the concepts of **beliefs** and **best response**
- Define and identify **Nash equilibria**
- Understand conditions for **unique** and **multiple** Nash equilibria
- Apply **selection procedures** among multiple Nash equilibria

# Outline

1. Beliefs and best response
2. Nash equilibrium
3. Finding Nash equilibria
4. Selection among multiple equilibria

1. **Beliefs and best response**

# Best response

We already mentioned this concept in Lecture 2.

**Definition (Best response strategy)**

Given a normal form game $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ and a strategy profile $s_{-i} \in S_{-i}$ (for the other players), the best response strategy $s_i$ of player $i$ is given by

$$s_i = \underset{s \in S_i}{\operatorname{argmax}}\ u(s, s_{-i})$$

We write $b_i(s_{-i})$ the set of best responses to $s_{-i}$ (there may be more than one).

For the **"battle of the sexes"**, we had the following matrix representation

| | | Parker | |
|---|---|---|---|
| | | O | F |
| Alex | O | $2, 1$ | $0, 0$ |
| | F | $0, 0$ | $1, 2$ |

**Example best responses:**

- *If* Alex *knew* that Parker was choosing F, the best response for Alex is F, $b_{\text{Alex}}(\text{F}) = \text{F}$
- *If* Alex *knew* that Parker was choosing O, the best response for Alex is O, $b_{\text{Alex}}(\text{O}) = \text{O}$

## Why is that important?

Remember that players are rational.

**Proposition (rational players and best response)**

*A rational player, who believes that the others players have chosen a strategy profile $s_{-i}$, will always choose their best response to $s_{-i}$ (or, one of them, if there is more than one).*

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This comes with a few related results and consequences...

**Proposition (strictly dominated strategies and best response)**

*A strategy $s_i$ of player $i$, if strictly dominated, can never be a best response to any of the opponents' strategy $s_{-i} \in S_{-i}$.*

**Proposition (equilibria and best response)**

*In a finite normal form game, if a strategy profile $s^*$*

- *is a strict dominant equilibrium, or if*
- *it uniquely survives the IESDS,*

*then for any player $i$ ($i = 1, \ldots, n$), their strategy $s_i^*$ is a best response to the opponents' strategy profile $s_{-i}^*$.*

# It is all about beliefs...

We have used the expression "*if... knew*" and the word "*believe*".

**Definition (belief)**

A *belief* of player $i$, about their opponents' strategy, is defined as a strategy profile $s_{-i} \in S_{-i}$, which player $i$ thinks they may choose.

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And, for each belief, player $i$ may find their best-response strategy, leading to the **best-response correspondence**.

**Definition (best-response correspondence)**

The *best-response correspondence* of player $i$ is a mapping which returns, for each potential strategy profile $s_{-i} \in S_{-i}$ of the other players, the corresponding set $b_i(s_{-i})$ of best response strategies, i.e.

$$b_i(s_{-i}) = \{s_i \in S_i \,:\, u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i}), \forall s_i' \in S_i\}$$

It is a like a look-up table (or cookbook) for player $i$ to decide what to play, given what they think others will be playing.

2. **Nash equilibrium**

# Rationality and common knowledge do not tell everything

So far, we have seen that

- strict dominance, based on rationality only, allows to predict the solution of some games (e.g., prisoner dilemma)
- IESDS (and rationalization), based on both rationality and common knowledge, allow to find the solutions of other games (it is a more general concept)

Though, for the example of the "battle of the sexes", we still do not have an approach to define a solution...

| | | Parker | |
|---|---|---|---|
| | | O | F |
| Alex | O | $2, 1$ | $0, 0$ |
| | F | $0, 0$ | $1, 2$ |

We need (even) more general solution concepts!

# The Nash equilibrium

Nash proposed an idea based on the "**concentration of best responses**".

**Definition (pure strategy Nash equilibrium)**

Given a normal form game $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$, the strategy profile $s^* = (s_1^*, \ldots, s_n^*)$ is a *pure strategy Nash equilibrium* of $\Gamma$ if

$$s_i^* \in b_i(s_{-i}^*), \quad \forall i$$

For the **"battle of the sexes"**,

- We have $b_{\text{Alex}}(\text{F}) = \text{F}$ and $b_{\text{Parker}}(\text{F}) = \text{F}$
- We also have $b_{\text{Alex}}(\text{O}) = \text{O}$ and $b_{\text{Parker}}(\text{O}) = \text{O}$

There are 2 Nash equilibria for this game!

| | | Parker | |
|---|---|---|---|
| | | O | F |
| Alex | O | $2, 1$ | $0, 0$ |
| | F | $0, 0$ | $1, 2$ |

# A pragmatic interpretation

Why is this considered as an equilibrium?

**Proposition (the pragmatic equilibrium)**

*For a normal form game* $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$*, considering any given pure strategy Nash equilibrium* $s^* = (s_1^*, \ldots, s_n^*)$*,* **not a single player** $i$ **could be better off by unilaterally deviating from their strategy** $s_i^*$.

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Consider the prisoner dilemma example:

| | | player 2 | |
|---|---|---|---|
| | | C | NC |
| player 1 | C | $-4, -4$ | $-1, -10$ |
| | NC | $-10, -1$ | $-2, -2$ |

We observe that,

- (C,C) is the only pure strategy Nash equilibrium
- none of the players have an interest in deviating from that solution

# Important results

The Nash equilibrium generalizes the type of equilibria we have seen before

**Proposition (a general equilibrium concept)**

*For a normal form game* $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$*, if a strategy profile* $s^* = (s_1^*, \ldots, s_n^*)$ *is*

- *a strict dominant strategy equilibrium, or*
- *a weak dominant strategy equilibrium, or*
- *an IESDS (or rationalization-based) equilibrium*

*then, it is also a pure Nash equilibrium.*

## Important results

The Nash equilibrium generalizes the type of equilibria we have seen before

**Proposition (a general equilibrium concept)**

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- *a strict dominant strategy equilibrium, or*
- *a weak dominant strategy equilibrium, or*
- *an IESDS (or rationalization-based) equilibrium*

*then, it is also a pure Nash equilibrium.*

And, we know an additional condition for which the Nash equilibrium is unique

**Proposition (uniqueness of Nash equilibrium)**

*For a normal form game $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$, if a strategy profile $s^* = (s_1^*, \ldots, s_n^*)$ is*

- *a strict dominant strategy equilibrium, or*
- *the unique survivor of IESDS (or rationalization)*

*then $s^*$ is the unique Nash equilibrium.*

3. **Finding Nash equilibria**

# A working example

We can find all pure strategy Nash equilibria of a normal form game, if at least one exists

Consider this normal form game, with the following matrix representation:

| | | player 2 | | |
|---|---|---|---|---|
| | | L | C | R |
| player 1 | U | $7, 7$ | $4, 2$ | $1, 8$ |
| | M | $2, 4$ | $5, 5$ | $2, 3$ |
| | D | $8, 1$ | $3, 2$ | $0, 0$ |

There is no strictly dominated strategy, so, using rationality and common knowledge only, we cannot see any solution to that game...

# Identify highest pay-offs (1)

We want to find the maximum pay-offs for each player and each strategy.

For instance, here, we start by looking at maximum pay-offs for player 1, for each strategy potentially chosen by player 2 (and underlining them):

| | | player 2 | | |
|---|---|---|---|---|
| | | L | C | R |
| player 1 | U | $7, 7$ | $4, 2$ | $1, 8$ |
| | M | $2, 4$ | $\underline{5, 5}$ | $\underline{2, 3}$ |
| | D | $\underline{8, 1}$ | $3, 2$ | $0, 0$ |

This shows us the **best-response correspondence** of player 1

# Identify highest pay-offs (2)

We want to find the maximum pay-offs for each player and each strategy.

We continue, here, by looking at maximum pay-offs for player 2, for each strategy potentially chosen by player 1 (and overlining them):

| | | player 2 | | |
|---|---|---|---|---|
| | | L | C | R |
| player 1 | U | $7,7$ | $4,2$ | $\overline{1,8}$ |
| | M | $2,4$ | $\overline{5,5}$ | $\underline{2,3}$ |
| | D | $\underline{8,1}$ | $\overline{3,2}$ | $0,0$ |

This shows us the **best-response correspondence** of player 2

# Concentration of best responses

As originally explained, a Nash equilibrium is a concentration of best responses.

Hence, any entry that is both overlined and underlined is a Nash equilibrium(!)

| | | player 2 | | |
|---|---|---|---|---|
| | | L | C | R |
| player 1 | U | $7,7$ | $4,2$ | $\overline{1,8}$ |
| | M | $2,4$ | $\overline{\underline{5,5}}$ | $\underline{2,3}$ |
| | D | $\underline{8,1}$ | $\overline{3,2}$ | $0,0$ |

**The Nash equilibrium is (M,C)**

4. **Selection among multiple equilibria**

# A working example

**It is very often that one finds multiple Nash equilibria in Normal form games.**
(and more generally for other types of games)

Consider this game (called a *coordination game*), with the following matrix representation:

| | | player 2 | |
|---|---|---|---|
| | | A | B |
| player 1 | C | $12, 12$ | $0, 10$ |
| | D | $10, 0$ | $8, 8$ |

There are **2 Nash equilibria**, i.e. (C,A) and (D,B)

# Pay-off dominant Nash equilibrium

The **pay-off dominant Nash equilibrium** is that which gives **highest pay-offs for all players overall**.

**Definition (pay-off dominant Nash equilibrium)**

For a normal form game $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ with multiple Nash equilibria $s^{j*} = (s_1^{j*}, \ldots, s_n^{j*})$, $j = 1, \ldots, J$, the *pay-off dominant Nash equilibrium* is the one that is *Pareto superior* to all others.
An equilibrium is *Pareto superior* if it improves the pay-off of at least 1 player, without making others worse off.

| | | player 2 | |
|---|---|---|---|
| | | A | B |
| player 1 | C | 12, 12 | 0, 10 |
| | D | 10, 0 | 8, 8 |

In this game, (C,A) is the **pay-off dominant Nash equilibrium**, since giving higher pay-offs than (D,B), to both players.

# Risk dominant Nash equilibrium

The **risk dominant Nash equilibrium** is that which minimizes the **joint exposure of all players** (i.e., the potential loss if deviating from the equilibrium).

**Definition (risk dominant Nash equilibrium)**

For a normal form game $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ with multiple Nash equilibria $s^{j*} = (s_1^{j*}, \ldots, s_n^{j*})$, $j = 1, \ldots, J$, the *risk dominant Nash equilibrium* is the one that minimizes the product of the loss exposure of all players, if they unilaterally deviate from their equilibrium strategy.

player 2

| player 1 | | A | B |
|---|---|---|---|
| | C | $12, 12$ | $0, 10$ |
| | D | $10, 0$ | $8, 8$ |

In this game, (C,A) is (also) the **risk dominant Nash equilibrium**, since

- for (D,B), the product of loss exposure for both players is $(8 - 0) \times (8 - 0) = 64$
- for (C,A), the product of loss exposure for both players is $(12 - 10) \times (12 - 10) = 4$

# Wrapping up

- The **Nash equilibrium** is deemed the most important solution concept in game theory
- It relies on the determination of the **best-response correspondence** and on the **concentration of best responses**
- **There does not always exist** a Nash equilibrium, and **if so, it is often not unique**
- **Selecting among equilibria** requires additional concepts and can end up being fairly complex...

**Further readings (/lecture notes for this topic):**

Tadelis S (2013) *Game Theory - An Introduction* - Chap. 5 (available on Blackboard Learn)
Narahari Y (2014) *Game Theory and Mechanism Design* - Chap. 6 (available on Blackboard Learn)

**Thanks for your attention!**